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Muscle actuator system

Abstract

A muscle actuator includes: a fitting; an inner tube; a ferrule; an outer sleeve; and a potting material. The inner tube defines a proximal section arranged within the fitting. The needle is arranged within the proximal section of the inner tube and extending through the fitting. The ferrule: is arranged about the proximal section of the inner tube within the fitting; and compresses the proximal section of the inner tube about the needle. The outer sleeve: concentrically encompasses the inner tube; is configured to increase in diameter response to increased pressure within the inner tube; and defines a proximal sleeve section defining a void within the fitting. The potting material is configured to: occupy the void within the fitting; retain the inner tube, the needle, the ferrule, and the outer sleeve against the fitting; and transfer tensile forces from the inner tube and the outer sleeve into the fitting.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This Application claims the benefit of U.S. Provisional Application Nos. 63/531,532, filed on 8 Aug. 2023, and 63/531,533, filed on 8 Aug. 2023, each of which is hereby incorporated in its entirety by this reference.

TECHNICAL FIELD

(1) This invention relates generally to the field of actuators and more specifically to a new and useful system for a high-pressure muscle actuator in the field of actuators.

Description

BRIEF DESCRIPTION OF THE FIGURES

- (1) FIG. 1 is a schematic representation of a system;
- (2) FIG. 2 is a schematic representation of the system;
- (3) FIG. 3 is a schematic representation of the system;
- (4) FIG. 4 is a schematic representation of the system;
- (5) FIG. 5 is a schematic representation of the system; and
- (6) FIGS. 6A and 6B are schematic representations of the system.

DESCRIPTION OF THE EMBODIMENTS

(7) The following description of embodiments of the invention is not intended to limit the invention to these embodiments but rather to enable a person skilled in the art to make and use this invention. Variations, configurations, implementations, example implementations, and examples described herein are optional and are not exclusive to the variations, configurations, implementations, example implementations, and examples they describe. The invention described herein can include any and all permutations of these variations, configurations, implementations, example implementations, and examples.

1. System

(8) As shown in FIGS. 1 and 2, a system **100** includes: a fluid distribution block **110**; a fitting **120**; an inner tube **130**; a needle **140**; a ferrule **150**; an outer sleeve **160**; and a volume of potting material **170**.

(9) The fitting **120**: defines a cavity **121**; defines a bore **122** extending from a proximal end of the fitting **120** to the cavity **121**; includes a set of retention features **123** arranged about an inner wall of the cavity **121**; and is coupled to the fluid distribution block **110**.

(10) The inner tube **130** defines a proximal tube section arranged within the cavity **121** of the fitting **120**.

(11) The needle **140**: is arranged within the proximal tube section of the inner tube **130** and extends through the bore **122** of the fitting **120**; and is configured to transfer a fluid from the fluid distribution block **110** through the inner tube **130** to increase pressure within the inner tube **130**.

(12) The ferrule **150**: is arranged about the proximal tube section of the inner tube **130** within the cavity **121**; compresses the proximal tube section of the inner tube **130** about the needle **140**; and is offset from a proximal end of the inner tube **130**.

(13) The outer sleeve **160**: concentrically encompasses the inner tube **130**; defines a nominal sleeve diameter configured to increase responsive to increased pressure within the inner tube **130**; and defines a proximal sleeve section forming an outer annular void **163** between the inner wall of the cavity **121** and an outer surface of the outer sleeve **160** and an inner annular void **162** between the needle **140** and an inner surface of the outer sleeve **160**.

(14) The volume of potting material **170** is arranged within the cavity **121** of the fitting **120** and configured to: occupy the inner annular void **162**, the outer annular void **163**, and the set of retention features **123** within the cavity **121** of the fitting **120**; retain the proximal tube section of the inner tube **130**, the needle **140**, the ferrule **150**, and the proximal sleeve section of the outer sleeve **160** within the fitting **120**; and transfer tensile forces from the outer sleeve **160** into the fitting **120** and the fluid distribution block **110**.

(15) The outer sleeve **160** is operable in: a retracted position, wherein the outer sleeve **160** retracts to the nominal sleeve diameter, responsive to supply of fluid at a first fluid pressure to the inner

tube **130** by the fluid distribution block **110**, in the retracted position; and an expanded position, wherein the outer sleeve **160** expands to a target sleeve diameter, responsive to supply of fluid at a second fluid pressure to the inner tube **130** by the fluid distribution block **110** that expands the inner tube **130** and the outer sleeve **160** and increases tensile forces applied by the outer sleeve **160** onto the fluid distribution block **110**, in the expanded position. The target sleeve diameter greater than the nominal sleeve diameter, and the second fluid pressure greater than the first fluid pressure.

2. Applications

(16) Generally, the system **100** functions as a compact, lightweight, actuator operable in a retracted configuration and an expanded configuration in order to retract (or “pull”) and extend a load coupled to the muscle actuator. More specifically, the system **100** includes an outer sleeve **160** (e.g., braided sleeve) configured to operate: in a retracted position responsive to supply of a nominal flow rate of a high-pressure fluid into an inner tube **130** (or “bladder”) arranged within the outer sleeve **160**; and an expanded position responsive to supply of a target flow rate, greater than the nominal flow rate, of the high-pressure fluid into the inner tube **130** to 1) increase a diameter of the outer sleeve **160** from a nominal sleeve diameter, 2) decrease a length of the outer sleeve **160** from a nominal sleeve length, and 3) transfer tensile forces from the outer sleeve **160** into a fluid distribution block **110** coupled to the inner tube **130** and the outer sleeve **160** to pull a load coupled to a proximal end of the inner tube **130**. Accordingly, rather than a pull force of the load constrained to a fixed diameter of a cylindrical structure, the system **100** can output a greater pull force per unit fluid pressure according to: a fluid pressure within the inner tube **130**; and a variable diameter of the outer sleeve **160**.

(17) The system **100** further includes a fitting **120**: defining a cavity **121** configured to receive a proximal end of the inner tube **130**; defining a bore **122** arranged within the fitting **120** and connecting an inlet port **124** at a proximal end of the fitting **120** to the cavity **121**; and including a set of retention features **123** (e.g., undercuts) arranged about an inner wall of the cavity **121**. The system **100** further includes a needle **140** (e.g., rigid tubular structure): arranged within a proximal end of the inner tube **130** and extending through the bore **122** within the fitting **120**; and configured to transfer the high-pressure fluid supplied from the fluid distribution block **110** into an interior volume of the inner tube **130**. The system **100** also includes a ferrule **150** (e.g., crimped ferrule): arranged about the inner tube **130**; offset the proximal end of the inner tube **130**; and compressing the inner tube **130** to retain the inner tube **130** about the needle **140** and define a transition region between an outer surface of the inner tube **130** and the ferrule **150**.

(18) The outer sleeve **160** is arranged about the inner tube **130** and extends within the cavity **121** of the fitting **120** to define: an outer annular void **163** between the inner wall of the cavity **121** and an outer surface of the outer sleeve **160**; and an inner annular void between the needle **140** and an inner surface of the outer sleeve **160**. Accordingly, the system **100** can then implement potting techniques (e.g., such as pouring, vacuum potting, and injection potting) to inject a volume of potting material **170** within the cavity **121** to occupy the inner annular void **162** and the outer annular void **163**. The potting material **170** is configured to: retain the inner tube **130**, the needle **140**, the ferrule **150**, and the outer sleeve **160** against the fitting **120**; and transfer tensile forces from the outer sleeve **160**—resulting from increased pressure in the inner tube **130**—into the fitting **120** and therefore, into the fluid distribution block **110**.

(19) Therefore, the system can function as a high-pull force actuator, with minimal components, configured to output a target pull force according to: an internal pressure within the inner tube **130**; a variable diameter of the outer sleeve **160**; and twist directions braided strands (e.g., fibers) of the outer sleeve.

2.1 Applications: Internal Fluid Seal

(20) The system **100** can trigger (e.g., via a controller) the fluid distribution block **110** to: supply the high-pressure fluid into an interior volume of the inner tube **130**; expand the inner tube **130** responsive to increased pressure within the inner tube **130**; and expand the outer sleeve **160** to

transfer tensile forces into the fitting **120**. In this example, during supply of the high-pressure fluid into the interior volume of the inner tube **130**, the high-pressure fluid can leak (or “seep”) into the cavity **121** of the fitting **120**. In particular, the high-pressure fluid can flow into: an inner interstice at the inner annular void **162** between an inner surface of the inner tube **130** and an outer surface of the needle **140**; and an outer interstice at the outer annular void **163** between the inner wall of the cavity **121** and the potting material **170** occupying the outer annular void **163**.

(21) The high-pressure fluid flowing through the inner interstice results in a concentration of stress at the transition region between the inner tube **130** and the ferrule **150** that forms a ramp **132** over the proximal end of the ferrule **150** functioning as an inner fluid seal configured to prevent the high-pressure fluid from flowing to an external environment. Additionally, the high-pressure fluid flowing through the outer interstice traverses through the set of retention features **123**, about the inner wall of the cavity **121**, which functions as an outer fluid seal, cooperating with the inner fluid seal, to prevent the high-pressure fluid from flowing to the external environment.

(22) Therefore, rather than implementing additional fluid seal components (e.g., O-rings, end cap seals, diaphragm seals) into the fitting **120** to prevent the high-pressure fluid leaking from the fitting **120**, the system **100** includes the potting material **170**: cooperating with a transition region between the inner tube **130** and the ferrule **150** to form a primary fluid seal for the fitting **120**; and cooperating with the retention features in the cavity **121** to form a secondary fluid seal for the fitting **120**. Accordingly, the system **100** can then trigger the fluid distribution block **110** to: regulate (e.g., increase, decrease) flow rate of the high-pressure fluid into the inner tube **130**; and transition between an extended position and a retracted position of the load coupled to the fluid distribution block **110**.

2.2 Applications: Abrasion Resistance

(23) The system **100** can further include an intermediate tube **134**: interposed between the inner tube **130** and the outer sleeve **160**; and configured to shield the inner tube **130** from abrasion and extrusion against the outer sleeve **160** during expansion and retraction of the inner tube **130** responsive to increased pressure within the inner tube **130**. In one example, the inner tube **130**: is formed of a low-durometer (e.g., between 20 and 40 Shore A) elastomeric material; and defines an inner tube **130** diameter. Additionally, the outer sleeve **160**: is formed of a high-durometer (e.g., greater than 60 Shore A) elastomeric material; defines an intermediate tube **134** diameter concentric with the inner tube **130** diameter of the inner tube **130**; and is interposed within a nominal gap between the inner tube **130** and the outer sleeve **160**.

(24) Therefore, during repeated expansion and contraction of the inner tube **130**, the intermediate sleeve **168** functions as a shield (or “wear plate”) to prevent abrasion between an outer surface of the inner tube **130** and an inner surface of the outer sleeve **160** and thus, prevent wear of the inner tube **130** during repeated (e.g., greater than one million cycles) retraction and expansion of the inner tube **130**.

3. Fitting

(25) Generally, as shown in FIGS. **1**, **2**, and **3**, the system **100** includes a fitting **120** functioning as an interface between an inner tube **130** and a fluid distribution block **110** (e.g., manifold) supplying a high-pressure fluid (3000 pounds per square inch) into the inner tube **130**. More specifically, the fitting **120**: includes a distal end coupled to a proximal section of an inner tube **130** (e.g., elastomeric tube) and an outer sleeve **160** (e.g., braided sleeve); includes a proximal end coupled to a fluid distribution block **110** (e.g., manifold) configured to supply a high-pressure fluid (e.g., synthetic fluid, mineral oil-based fluid) into the inner tube **130**; and is configured to transfer tensile forces (e.g., hoop stresses) about the inner tube **130** and the outer sleeve **160** into the fluid distribution block **110** responsive to increased pressure at the inner tube **130**.

(26) In one implementation, the fitting **120** can be formed of a rigid material (e.g., stainless steel, aluminum) and includes: a block-mounting section configured to couple to a fluid distribution block **110**; and a tube-mounting section configured to couple to the inner tube **130** and the outer

sleeve **160**. In this implementation, the block-mounting section defines a substantially tapered geometry and includes: an inlet port **124** arranged at a proximal end of the fitting **120** and configured to couple to an outlet port (e.g., high-pressure fluid outlet port) of the fluid distribution block **110**; a groove **125** inset from an outer surface of the block-mounting section and offset from the inlet port **124**; and a gasket **126** (e.g., elastomeric gasket) arranged about the groove **125** and configured to prevent flow of the fluid supplied from the fluid distribution block **110** across the outer surface of the fitting **120**. In one example, the block-mounting section also includes a set of threads **127**: arranged about the outer surface of the block section offset from the groove **125**; configured to retain the fitting **120** within the fluid distribution block **110**; and configured to transfer tensile forces from the inner tube **130** and the outer sleeve **160** to the fluid distribution block **110** responsive to increase in fluid pressure at the inlet port **124**.

(27) In this implementation, the tube-mounting section defines a substantially cylindrical geometry and includes: a cavity **121** (e.g., cylindrical volume) arranged at a distal end of the fitting **120** and configured to receive proximal ends of the inner tube **130** and the outer sleeve **160**; a bore **122** (e.g., cylindrical channel) extending from the inlet port **124** at the proximal end of the fitting **120** to the cavity **121** and the distal end of the fitting **120**; and a set of retention features **123** (e.g., undercuts) arranged about an inner wall of the cavity **121** configured to prevent flow of high-pressure fluid-leaked within the cavity **121**—to an outside environment. In one example, the set of retention features **123** extend from a distal end of the fitting **120** to a section (e.g., a midpoint) of the cavity **121** offset from the proximal end of the fitting **120**. In this example, each retention feature, in the set of retention features **123**: includes an undercut formed inset from the inner wall of the first cavity; and defines a triangular geometry.

(28) Therefore, the system **100** includes the fitting **120**: to function as an interface between the inner tube **130** and the fluid distribution block **110**; to function as a liquid seal to direct a high-pressure fluid into the inner tube **130**; and to function as a load-bearing element to transfer tensile forces across the outer sleeve **160** into the fluid distribution block **110** during operation of the system **100**.

3.1 Needle

(29) In one implementation, as shown in FIG. 2, the system **100** further includes a needle **140** (e.g., hollow-rigid tubing): arranged within the bore **122** of the fitting **120**; extending from the inlet port **124** at the proximal end of the fitting **120** into the cavity **121** at the distal end of fitting; inserted within the inner tube **130** at the cavity **121**; and configured to transfer a high-pressure fluid supplied at the fluid distribution block **110** into the inner tube **130**. In this implementation, the needle **140** is inserted within the inner tube **130** such that a proximal end of the inner tube **130** is arranged offset (e.g., 0.25 inch offset) from a proximal end of the needle **140** in order to locate the proximal tube section entirely within the cavity **121** of the fitting **120**.

(30) Thus, the needle **140** functions as an interface to transfer a high-pressure fluid supplied from the fluid distribution block **110** into the proximal tube section of the inner tube **130** arranged within the tube-mounting section of the fitting **120**.

4. Inner Tube

(31) In one implementation, as shown in FIGS. 2 and 3, the system **100** includes an inner tube **130**: formed of a low-durometer (e.g., between 20 and 40 Shore A) elastomeric material; defining a nominal tube length (e.g., four inches); defining a proximal tube section configured to couple within a cavity **121** of a primary fitting **120**; and defining a distal tube section configured to couple within a secondary cavity of a secondary fitting. As described above, the proximal tube section: is coupled to the needle **140** within the cavity **121** of the fitting **120**; and is configured to receive a high-pressure fluid into an interior volume of the inner tube **130** from the needle **140**.

(32) In this implementation, the primary fitting **120** can be coupled to a primary fluid distribution block and the secondary fitting can be coupled to a secondary fluid distribution block and/or a mounting block **112** in order to form a trapped pressure volume within the inner tube **130**.

Accordingly, the inner tube **130**: defines a nominal tube diameter (e.g., 0.08 inches) responsive to supply of fluid at a baseline fluid pressure (e.g., less than 3000 pounds per square inch) into the inner tube **130** from the fluid distribution block **110**; and defines a target tube diameter (e.g., 0.099 inches) responsive to supply of fluid at a target fluid pressure (e.g., greater than 3000 pounds per square inch) into the inner tube **130** from the fluid distribution block **110**. Thus, as the diameter of the inner tube **130** transitions from the nominal tube diameter to the target tube diameter, the nominal tube length (e.g., four inches) decreases to a target tube length (e.g., 3.5 inches).

(33) Therefore, the inner tube **130** can function as a high-pressure (e.g., greater than 3000 pounds per square inch) bladder configured to operate in a retracted position and an expanded position responsive to supply of fluid at high-pressure fluid into the inner tube **130** by the fluid distribution block **110**.

4.1 Crimped Ferrule

(34) In one implementation, as shown in FIGS. 2 and 5, the system **100** includes a ferrule **150** (e.g., crimped ferrule): arranged about the proximal tube section of the inner tube **130** within the cavity **121** of the fitting **120**; and compressing the proximal tube section of the inner tube **130**, offset from a proximal end of the inner tube **130**, against the needle **140**. In particular, the ferrule **150** defines a substantially cylindrical structure that is crimped about the proximal tube section to form a crimped section and an uncrimped section. In this implementation, the crimped section: is crimped about the proximal tube section to define a first ferrule **150** diameter concentric about a diameter of the proximal tube section of the inner tube **130**; compresses the proximal tube section of the inner tube **130** against an outer surface of the needle **140** in order to form concentration of stress at a transition region between an outer surface of the inner tube **130** and the ferrule **150**; and retains the needle **140** coupled to the inner tube **130** to supply flow of high-pressure fluid into the inner tube **130** by the fluid distribution block **110**.

(35) In one example, crimping techniques—such as rotary crimping, press crimping, electric crimping, hand crimping—can be implemented to form the crimped section and thus: define a first ferrule **150** diameter substantially approximating (e.g., ± 0.001 inches) the nominal tube diameter; and form a press fit (or interference fit) of the ferrule **150** about the inner tube **130** to prevent flow of fluid through an interstice between the needle **140** and an inner surface of the inner tube **130** at the proximal tube section.

(36) More specifically, the concentration of stress at the transition region forms a ramp **132** of the proximal tube section over a proximal end of the crimped section of the ferrule **150** configured to prevent flow of fluid through an interstice between the ramp **132** and the ferrule **150**.

(37) Additionally, the uncrimped section: defines a second ferrule **150** diameter, greater than the first ferrule **150** diameter, concentric about the diameter of the proximal tube section of the inner tube **130**; forms a nominal gap between the inner tube **130** and the outer sleeve **160** enclosing the inner tube **130**; and terminates proximal a distal end of the fitting **120** to locate the ferrule **150** entirely within the cavity **121** of the fitting **120**. Accordingly, the inner tube **130** is configured to, responsive to supply of a target fluid pressure by the fluid distribution block **110** into the inner tube **130**: expand to reduce the nominal gap between the inner tube **130** and the outer sleeve **160**; expand a nominal sleeve diameter of the outer sleeve **160** to a target sleeve diameter; and transfer tensile forces from the outer sleeve **160** into the fitting **120**.

(38) Therefore, the ferrule **150**: retains the inner tube **130** compressed against the needle **140** within the cavity **121** of the fitting **120**; maintains a nominal gap between the inner tube **130** and the outer sleeve **160**; and forms a ramp **132** of the proximal tube section over a proximal end of the ferrule **150** to function as a gasket to prevent flow of liquid through an interstice between the ramp **132** and the ferrule **150**.

4.2 Intermediate Tube

(39) In one implementation, as shown in FIGS. 1 and 2, the system **100** further includes an intermediate tube **134**: interposed between the inner tube **130** and the outer sleeve **160**; and offset

from the proximal tube section of the inner tube **130**. As described above, the inner tube **130** is configured to expand to reduce the nominal gap between the inner tube **130** and the outer sleeve **160** responsive to supply of a target fluid pressure by the fluid distribution block **110** into the inner tube **130**. Accordingly, an outer surface of the inner tube **130** expands to contact an inner surface of the outer sleeve **160**, which results in abrasion of the outer surface of the inner tube **130** during repeated retraction and expansion of the inner tube **130**. Thus, the intermediate tube **134** is configured to prevent abrasion of the inner tube **130** during repeated retraction and expansion of the inner tube **130**.

(40) In one example, the inner tube **130**: is formed of a low-durometer (e.g., between 20 and 40 Shore A) elastomeric material; and defines an inner tube **130** diameter. Additionally, the outer sleeve **160**: is formed of a high-durometer (e.g., greater than 60 Shore A) elastomeric material; defines an intermediate tube **134** diameter concentric with the diameter of the inner tube **130**; is interposed within the nominal gap between the inner tube **130** and the outer sleeve **160**; and is configured to shield the inner tube **130** from extrusion through the outer sleeve **160** and abrasion from the outer sleeve **160** against the outer surface of the inner tube **130** during increased pressure within the inner tube **130**. Accordingly, the intermediate tube **134** prevents the inner tube **130** from impregnating interstices between fibers across an inner surface of the outer sleeve **160**—during expansion of the inner tube **130** within the outer sleeve—thereby, preventing abrasion between the inner tube **130** and the outer sleeve **60**.

(41) Therefore, the system **100** includes an intermediate sleeve **168**: arranged within a nominal gap between the inner tube **130** and the outer sleeve **160**; and functioning as a shield to prevent abrasion between an outer surface of the inner tube **130** and an inner surface of the outer sleeve **160** and thus, prevent wear of the inner tube **130** during repeated retraction and expansion of the inner tube **130**.

(42) The system **100** can implement the structure described above to form multiple (e.g., n-number) of intermediate tubes **134** interposed between the inner tube **130** and the outer sleeve **160** to shield the inner tube **130** from abrasion contact with the inner surface of the outer sleeve **160**.

5. Outer Sleeve+Annular Voids

(43) In one implementation, as shown in FIGS. 2 and 3, the system **100** includes an outer sleeve **160** (e.g., braided sleeve): concentrically encompassing the inner tube **130**; and defining a nominal sleeve diameter configured to expand responsive to expansion of the inner tube **130**, as described above, resulting from increased pressure within the inner tube **130**. In this implementation, the outer sleeve **160** includes a proximal sleeve section arranged within the cavity **121** of the fitting **120** to: form an outer annular void **163** between an inner wall of the cavity **121** and an outer surface of the outer sleeve **160**; and form an inner annular void **162** between the needle **140** and an inner surface of the outer sleeve **160**. Additionally, as described below, the system **100** can include a volume of potting material **170**: arranged within the cavity **121** to occupy the inner annular void **162** and the outer annular void **163**; and retain the inner sleeve and the outer sleeve **160** to the fitting **120**.

(44) Thus, in response to supplying a target fluid pressure within the inner tube **130** by the fluid distribution block **110**, the outer sleeve **160** transitions from a retracted position to an expanded position to transfer tensile forces across the outer sleeve **160** into the fitting **120** and thus, into the fluid distribution block **110**.

(45) The system **100** can implement the structure described above to form multiple (e.g., n-number) of additional sleeves (e.g., braided sleeves) interposed between the inner tube **130** and the outer sleeve **160** to increase pressure tolerances of the system **100** during retraction and expansion of the outer sleeve **160**. Accordingly, the outer sleeve can further define a distal sleeve section arranged within a secondary cavity of a secondary fitting and defining: a second outer annular void between an inner wall of the secondary cavity and the outer sleeve; and a secondary inner annular void between the needle and the outer sleeve.

5.1 Potting Material

(46) In one implementation, potting techniques—such as pouring, vacuum potting, and injection potting—can be implemented to locate a volume of potting material **170** within the cavity **121** to occupy the outer annular void **163** and the inner annular void **162** and thus, retain the inner tube **130** and the outer sleeve **160** within the cavity **121** of the fitting **120**. In one example, the volume of potting material **170** can be arranged within the cavity **121** by: injecting the potting material **170** into the outer annular void **163** such that, the potting material **170** flows through the outer annular void **163** and into the inner annular void **162** to fill the cavity **121** of the fitting **120**; and curing the potting material **170** following a target duration of time.

(47) More specifically, the potting material **170** flows into the outer annular void **163** to: impregnate interstices between fibers across the outer surface of the outer sleeve **160** to bond the outer sleeve **160** to the potting material **170** and thus, retain the outer sleeve **160** within the cavity **121** of the fitting **120**; and fill the set of retention features **123** (i.e., fill undercuts formed into the inner wall of the cavity **121**) within the fitting **120**. In one example, the potting material **170** occupies the outer annular void **163** to form a radiused transition **172** (e.g., a fillet): extending from a distal end of the fitting **120** to an outer surface of the outer sleeve **160**; and configured to reduce a stress concentration proximal the distal end of the fitting **120** during expansion of the outer sleeve **160**. Therefore, the radiused transition **172** functions as a buffer zone extending from the distal end of the fitting **120** to an outer surface of the outer sleeve **160** that reduces stress concentrations at the distal end of the fitting.

(48) Additionally, the potting material **170** flows into the inner annular void **162** to: impregnate interstices between fibers across the inner surface of the outer sleeve **160** to bond the outer sleeve **160** to the potting material **170** and retain the outer sleeve **160** within the cavity **121** of the fitting **120**; retain the needle **140**, the inner tube **130**, and the ferrule **150** within the cavity **121** of the fitting **120**; and cooperate with the ferrule **150** to form the concentration of stress at the transition region between an outer surface of the inner tube **130** and the distal end of the ferrule **150**.

(49) Therefore, the system **100** includes a volume of potting material **170**: occupying the cavity **121** of the fitting **120** to retain the inner tube **130** and the outer sleeve **160** within the fitting **120**; and functioning as a load-bearing element configured to transfer tensile forces across the outer sleeve **160**—from retraction and expansion of the outer sleeve **160**—into the fitting **120** and therefore, transfer these tensile forces into the fluid distribution block **110**.

5.2 Intermediate Sleeve

(50) In one implementation, as shown in FIGS. 2 and 3, the system **100** can include an intermediate sleeve **168** (e.g., aramid, carbon fiber, fiberglass): arranged about the proximal sleeve section of the outer sleeve **160**; and cooperating with the potting material **170** to bond the outer sleeve **160** to the potting material **170** and to retain the outer sleeve **160** within the cavity **121** of the fitting **120**. In this implementation, the outer sleeve **160** includes a primary set of strands **164**: formed of a substantially non-abrasive material (e.g., liquid crystal polymer); and defining a primary surface roughness (e.g., between 4 and 16 microinches). For example, the set of strands can include an elastomeric coating **167** (e.g., acrylic coating, polyurethane coating, silicone coating, butyl coating) to define the primary surface roughness across the intermediate sleeve **168**.

(51) Additionally, the intermediate sleeve **168** includes a secondary set of strands **165**: interposed between the proximal sleeve section of the outer sleeve **160** and the inner wall of the cavity **121** of the fitting **120**; formed of an abrasive material (e.g., aramid, carbon fiber, fiberglass); and defining a secondary surface roughness (e.g., between 1000 and 2000 microinches) greater than the primary surface roughness. During injection of the potting material **170** into the outer annular void **163** as described above, rather than the potting material **170** impregnating interstices between fibers across the outer surface of the outer sleeve **160**, the potting material **170** impregnates interstices between fibers across an outer surface of the intermediate sleeve **168**, thereby bonding the intermediate sleeve **168** and the outer sleeve **160** to the potting material **170** and retaining the intermediate

sleeve **168** and the outer sleeve **160** within the cavity **121** of the fitting **120**.

(52) Therefore, the system **100** can include the intermediate sleeve **168**: interposed between the outer sleeve **160** and the inner wall of the cavity **121** of the fitting **120**; and functioning as a mechanical bonding agent (or “surface primer”) cooperating with the potting material **170** injected within the outer annular void **163** to retain the outer sleeve **160** within the cavity **121** of the fitting **120**.

5.2 Braids+Coating

(53) In one implementation, as shown in FIG. **4**, the outer sleeve **160** includes a braided structure **166** (e.g., diamond braid, biaxial braid, helical braid): including a set of braided strands forming a substantially cylindrical configuration about the inner tube **130**; and configured to transfer tensile forces across the outer sleeve **160**—responsive to increased pressure within the inner tube **130**—in an axial direction toward the fitting **120** and into the fluid distribution block **110**. In this implementation, the braided strands, in the set of braided strands, can be formed of metallic, non-polymer, liquid crystal polymers, and non-organic fibers, strands, wire etc.

(54) In one example, the outer sleeve **160** includes: a primary set of strands **164**; and a secondary set of strands **165** cooperating with the primary set of strands **164** to form a helical braided structure **166**. The primary set of strands **164**: includes groups of fibers twisted in a primary direction (e.g., Z-twist, 45-degrees); and are braided into a primary helical pattern according to a direction orthogonal (e.g., 135-degrees) to the primary direction of the groups of fibers. The secondary set of strands **165**: includes groups of fibers twisted in a secondary direction (e.g., S-twist, -45 degrees); are braided into a secondary helical pattern according to a direction orthogonal (e.g., 45-degrees) to the secondary direction of the groups of fibers; and cooperates with the primary set of strands **164** to form the helical braided structure **166** enclosing the inner tube **130**.

(55) In this example, each strand—in the primary set of strands **164** and the secondary set of strands **165**—can include an elastomeric coating **167** (e.g., acrylic coating, polyurethane coating, silicone coating, butyl coating) configured to: retain the strands in the helical braided structure **166** across a target length of the outer sleeve **160**; reduce friction between adjacent strands to increase flexibility of the outer sleeve **160** during expansion of the outer sleeve **160** resulting from increased pressure in the inner tube **130**; and defining the primary surface roughness (e.g., between 4 and 16 microinches), as described above.

(56) Accordingly, during expansion of the outer sleeve **160** responsive to increased pressure in the inner tube **130**, the orthogonal relationship between the primary set of strands **164** and the primary helical pattern and the orthogonal relationship between the secondary set of strands **165** and the secondary helical pattern cooperate to direct (or “guide”) tensile forces in an axial direction across the outer sleeve **160** into the fitting **120** and thus, into the fluid distribution block **110**.

(57) Therefore, the system **100** can include a braided outer sleeve **160**: enclosing the inner tube **130**; and functioning as a tension guide to direct tensile forces across the outer sleeve **160**—resulting from hoop stresses about the outer sleeve **160** during increased pressure within the inner tube **130**—in an axial direction toward the fitting **120** and thus, into the fluid distribution block **110**.

6. Leak Path+Seal

(58) In one implementation, as shown in FIG. **3**, the system **100** can trigger the fluid distribution block **110** to direct a high-pressure fluid (e.g., 3000 pounds per square inch) through the needle **140** within the fitting **120** and into the inner tube **130** in order to increase pressure within the inner tube **130** and thus, transition the outer sleeve **160** from a retracted position and into an expanded position. In this implementation, during flow of the high-pressure fluid into the inner tube **130**, the high-pressure fluid can leak (or “seep”) from an interior volume of the inner tube **130** and into the cavity **121** of the fitting **120**. Accordingly, the system **100** implements the structure and assembly methods described above to prevent flow (or “seeping”) of high-pressure fluid from the cavity **121** of the fitting **120** to an outside environment, which can result in failed operation of the system **100**.

(59) During operation of the system **100**, the high-pressure fluid can leak into the cavity **121** of the fitting **120**: through a primary interstice between an outer surface of the needle **140** and an inner surface of the inner tube **130**; and into the inner annular void **162** and the outer annular void **163** within the cavity **121**. In particular, the high-pressure fluid flows into an inner interstice—within the inner annular void **162**—between an inner section of potting material **170** adjacent the transition region between the inner tube **130** and the ferrule **150**. This flow of high-pressure fluid through the primary interstice cruises a concentration of stress into the transition region that cooperates with applied pressure to the inner tube **130** by the ferrule **150** to form the ramp **132** over the proximal end of the ferrule **150**. Accordingly, the ramp **132** can then function as a gasket to prevent flow of fluid past the transition region between the inner tube **130** and the ferrule **150** and thus, prevent flow (or “seeping”) of fluid to an outside environment.

(60) Additionally, the high-pressure fluid flows into an outer interstice—within the outer annular void **163**—between an outer section of potting material **170** and the inner wall of the cavity **121** of the fitting **120**. As the high-pressure fluid material flows through the outer interstice toward the distal end of the fitting **120**, the high-pressure fluid occupies the set of retention features **123** (e.g., undercuts) along the inner wall of the cavity **121**, which prevents the high-pressure fluid from leaking to the outside environment through the outer annular void **163**.

(61) Therefore, the system **100** includes: an inner fluid seal (or “gasket”) within the inner annular void **162** of the cavity **121** formed by concentration of stresses resulting from flow of high-pressure fluid into the inner annular void **162** cooperating with the transition region between the inner tube **130** and the ferrule **150**; and an outer fluid seal (or “gasket”) within the outer annular void **163** of the cavity **121** formed by the set of retention features **123** about the inner wall of the cavity **121** cooperating with the outer section of potting material **170** within the outer annular void **163**. Accordingly, the inner fluid seal and the outer fluid seal cooperate to prevent high-pressure fluid, seeping into the cavity **121** of the fitting **120**, from flowing to an outside environment.

7. Single-Acting Actuator

(62) In one implementation, as shown in FIGS. **1**, **6A**, and **6B**, the system **100** includes: a primary fitting **120** configured to couple the proximal tube section of the inner tube **130** to a fluid distribution block **110**; and a secondary fitting configured to couple a distal tube section of the inner tube **130** to a mounting block **112**. More specifically, the system **100** can implement the structure and assembly techniques described above to: retain proximal ends of the inner tube **130** and the outer sleeve **160** within the primary fitting **120**; and retain distal ends of the inner tube **130** and the outer sleeve **160** within the secondary fitting.

(63) However, rather than including a needle **140** within the secondary fitting to supply the high-pressure fluid into the interior volume of the inner tube **130**, the system **100** includes a stopper: arranged within the distal tube section of the inner tube **130** and extending through a secondary bore of the secondary fitting; and forming a seal configured to block flow of the high-pressure fluid, within the interior volume of the inner tube **130**, outwardly from a distal end of the inner tube **130**.

(64) In this implementation, the primary fitting **120** is coupled to the fluid distribution block **110**: including a primary coupling element (e.g., eyelet) configured to mechanically couple the fluid distribution block **110** to a load; and including an outlet port configured to supply the high-pressure fluid in a first direction into the interior volume of the inner tube **130** via the proximal end of the inner tube **130**. Additionally, the secondary fitting is coupled to the mounting block **112**: including a secondary coupling element (e.g., eyelet) configured to couple the mounting block **112** to a mechanical reference; and configured to prevent flow of the high-pressure fluid in a second direction, opposite the first direction, outwardly from the distal end of the inner tube **130**.

(65) The system **100** can then: trigger the fluid distribution block **110** to supply a nominal flow rate of high-pressure fluid in the first direction into the interior volume of the inner tube **130** to expand the outer sleeve **160** to a nominal sleeve diameter; and trigger the fluid distribution block **110** to

supply a target flow rate, greater than the nominal flow rate, of high-pressure fluid in the first direction into the interior volume of the inner tube **130** to expand the outer sleeve **160** to a target sleeve diameter, greater than the nominal sleeve diameter. Accordingly, the fluid distribution block **110** cooperates with the mounting block **112** to form a single-acting actuator. The system **100** can alternate between flow of the high-pressure fluid in the nominal flow rate and the target flow rate to expand and retract the outer sleeve **160** and thus, extend and retract the load coupled to the primary coupling element.

(66) Therefore, the system **100** can: selectively adjust flow of the high-pressure fluid in a single direction within the interior volume of the inner tube **130** at the proximal end of the inner tube **130** to adjust (or “regulate”) pressure within the interior volume of the inner tube **130** and thus, adjust applied tensile forces applied to the load coupled to the primary coupling element.

(67) Additionally, the system **100** can include an enclosure: configured to contain the inner tube **130** and the outer sleeve **160**; locating the primary coupling element at a proximal end of the enclosure; and locating the secondary coupling element at a distal end of the enclosure.

7.1 Double Acting Actuator

(68) In one implementation, the system **100** includes: a primary fitting **120** configured to couple the proximal tube section of the inner tube **130** to a primary fluid distribution block; and a secondary fitting configured to couple a distal tube section of the inner tube **130** to a secondary fluid distribution block. More specifically, the system **100** can implement the structure and assembly techniques described above to: retain proximal ends of the inner tube **130** and the outer sleeve **160** within the primary fitting **120**; and retain distal ends of the inner tube **130** and the outer sleeve **160** within the secondary fitting.

(69) In this implementation, the primary fitting **120** is coupled to a primary fluid distribution block: including a primary coupling element (e.g., eyelet) configured to mechanically couple the primary fluid distribution block to a load; and including a primary outlet port configured to supply a high-pressure fluid in a first direction into the interior volume of the inner tube **130** via the proximal end of the inner tube **130**. Additionally, the secondary fitting is coupled to a secondary fluid distribution block: including a secondary coupling element (e.g., eyelet) configured to mechanically couple the secondary fluid distribution block to a mechanical reference; and including a secondary outlet port configured to direct a high-pressure fluid within the interior volume of the inner tube **130** in a second direction, opposite the first direction, into the secondary fluid distribution block via the distal end of the inner tube **130**.

(70) The system **100** can then: trigger the primary fluid distribution block to supply the high-pressure fluid in the first direction into the interior volume of the inner tube **130** to expand the outer sleeve **160**; and trigger the secondary fluid distribution block to direct the high-pressure fluid in the second direction into the secondary fluid distribution block to retract the outer sleeve **160**.

Accordingly, the primary fluid distribution block cooperates with the secondary fluid distribution block to form a double-acting actuator to expand and retract the outer sleeve **160** and thus, extend and retract the load coupled to the primary coupling element.

(71) Therefore, the system **100** can selectively adjust flow of the high-pressure fluid in two directions within the interior volume of the inner tube **130** at proximal and distal ends of the inner tube **130** to adjust (or “regulate”) pressure within the interior volume of the inner tube **130** and thus, adjust applied tensile forces applied to the load coupled to the primary coupling element.

(72) Additionally, the system **100** can include an enclosure: configured to contain the inner tube **130** and the outer sleeve **160**; locating the primary coupling element at a proximal end of the enclosure; and locating the secondary coupling element at a distal end of the enclosure.

7.3 Variable Recruitment Actuation

(73) In one implementation, as shown in FIGS. **6A** and **6B**, the system **100** can include: a housing including a set of inlet ports **124** and configured to couple to a fluid distribution block **110**; a set of inner tubes **130** including a proximal end arranged within the housing and coupled to the set of

inlet ports **124** to receive a high-pressure fluid within an interior volume of each inner tube **130**, in the set of inner tubes **130**; and a set of outer sleeves **160** encircling the set of inner tubes **130** and configured to expand responsive to increased pressure within the set of inner tubes **130**. In this implementation, the system **100** can implement the structure and assembly techniques described above to retain proximal ends of the set of inner tubes **130** and the set of outer sleeve **160s** within the housing.

(74) The system **100** can then: trigger the fluid distribution block **110** to supply a nominal flow rate of high-pressure fluid in the first direction into the interior volume of the set of inner tubes **130** to expand the set of outer sleeve **160s** to a nominal sleeve diameter; and trigger the fluid distribution block **110** to supply a target flow rate, greater than the nominal flow rate, of high-pressure fluid in the first direction into the interior volume of the set of inner tubes **130** to expand the set of outer sleeve **160s** to a target sleeve diameter, greater than the nominal sleeve diameter.

(75) Therefore, the housing, the set of inner tubes **130**, and the set of outer sleeve **160s** cooperate to form a variable recruitment actuator configured to transfer tensile forces across the set of outer sleeve **160s** into the housing and thus, into the fluid distribution block **110** coupled to the load.

(76) The systems and methods described herein can be embodied and/or implemented at least in part as a machine configured to receive a computer-readable medium storing computer-readable instructions. The instructions can be executed by computer-executable components integrated with the application, applet, host, server, network, website, communication service, communication interface, hardware/firmware/software elements of a user computer or mobile device, wristband, smartphone, or any suitable combination thereof. Other systems and methods of the embodiment can be embodied and/or implemented at least in part as a machine configured to receive a computer-readable medium storing computer-readable instructions. The instructions can be executed by computer-executable components integrated by computer-executable components integrated with apparatuses and networks of the type described above. The computer-readable medium can be stored on any suitable computer readable media such as RAMs, ROMs, flash memory, EEPROMs, optical devices (CD or DVD), hard drives, floppy drives, or any suitable device. The computer-executable component can be a processor but any suitable dedicated hardware device can (alternatively or additionally) execute the instructions.

(77) As a person skilled in the art will recognize from the previous detailed description and from the figures and claims, modifications and changes can be made to the embodiments of the invention without departing from the scope of this invention as defined in the following claims.

Claims

1. A system comprising: a first fluid distribution block; a first fitting: defining a first cavity; defining a first bore extending from a proximal end of the first fitting to the first cavity; comprising a first set of retention features arranged about an inner wall of the first cavity; and coupled to the first fluid distribution block; an inner tube defining a proximal tube section arranged within the first cavity of the first fitting; a first needle: arranged within the proximal tube section of the inner tube and extending through the first bore of the first fitting; and configured to transfer a fluid from the first fluid distribution block into the inner tube to increase pressure within the inner tube; a first ferrule: arranged about the proximal tube section of the inner tube within the first cavity; and compressing the proximal tube section of the inner tube about the first needle, the first ferrule offset from a proximal end of the inner tube; an outer sleeve: concentrically encompassing the inner tube; defining a nominal sleeve diameter configured to expand responsive to increased pressure within the inner tube; and defining a proximal sleeve section forming: a first outer annular void between the inner wall of the first cavity and an outer surface of the outer sleeve; and a first inner annular void between the first needle and an inner surface of the outer sleeve; and a first volume of potting material arranged within the first cavity of the first fitting and configured to:

occupy the first inner annular void, the first outer annular void, and the first set of retention features within the first cavity of the first fitting; retain the proximal tube section of the inner tube, the first needle, the first ferrule, and the proximal sleeve section of the outer sleeve within the first fitting; and transfer tensile forces from the outer sleeve to the first fitting, the first fitting transferring tensile forces from the first volume of potting material into the first fluid distribution block.

2. The system of claim 1, wherein the outer sleeve is operable in: a retracted position, the outer sleeve retracted to the nominal sleeve diameter, responsive to supply of fluid at a first fluid pressure to the inner tube by the first fluid distribution block, in the retracted position; and an expanded position, the outer sleeve expanded to a first sleeve diameter, responsive to supply of fluid at a second fluid pressure to the inner tube by the first fluid distribution block that expands the inner tube and the outer sleeve and increases tensile forces applied by the outer sleeve onto the first fluid distribution block, in the expanded position, the first sleeve diameter greater than the nominal sleeve diameter, and the second fluid pressure greater than the first fluid pressure.

3. The system of claim 1: further comprising an intermediate tube interposed between the inner tube and the outer sleeve, the intermediate tube offset from the proximal tube section of the inner tube; and wherein the first ferrule comprises a crimped ferrule comprising: an uncrimped section enclosing the intermediate tube and the inner tube; a crimped section compressing the proximal tube section against an outer surface of the first needle to: form a seal configured to prevent flow of the fluid through an interstice between the outer surface of the first needle and the proximal tube section; and form a ramp of the proximal tube section over a proximal end of the crimped ferrule, the ramp configured to prevent flow of the fluid through the interstice.

4. The system of claim 1, wherein the first fluid distribution block is configured to transfer the fluid through the first needle into the inner tube, the fluid: flowing through a first interstice between the inner tube and an outer surface of the first needle; flowing through a second interstice between a first section of potting material, within the first inner annular void, and the proximal tube section, the proximal tube section: forming a ramp over a proximal end of the first ferrule from compression of the first ferrule, the first section of potting material, and the fluid against the proximal tube section; and configured to prevent flow of the fluid between the inner tube and the first ferrule; and flowing through a third interstice between the inner wall of the first cavity and a second section of potting material within the first outer annular void, the first set of retention features arranged about the inner wall and configured to prevent flow of the fluid to a distal end of the first fitting.

5. The system of claim 1, wherein the volume of potting material is formed by: injecting the potting material into the outer annular void, the potting material flowing through the outer annular void into the inner annular void; and curing the potting material.

6. The system of claim 1: wherein the set of retention features extends from a distal end of the first fitting to a first section of the cavity, offset from the proximal end of the first fitting, each retention feature, in the first set of retention features: comprising an undercut formed inset from the inner wall of the first cavity; and defining a triangular geometry; and wherein the first volume of potting material is configured to: occupy each undercut, of each retention feature, in the set of retention features; and prevent flow of the fluid through an interstice between a first section of potting material, within the first outer annular void, and the inner wall of the first cavity.

7. The system of claim 1, wherein the first fitting further comprises: an inlet port arranged at a proximal end of the first fitting, the first needle extending through the inlet port; a groove offset from the inlet port and inset from an outer surface of the first fitting; a gasket arranged about the groove and configured to prevent flow of the fluid from the first fluid distribution block across the outer surface of the first fitting; and a set of threads: arranged about an outer surface of the first fitting and offset from the groove; configured to retain the first fitting within the first fluid distribution block; and configured to transfer tensile forces from the inner tube and the outer sleeve to the first fluid distribution block responsive to a first fluid pressure at the inlet port increasing

pressure within the inner tube.

8. The system of claim 1: further comprising: a mounting block; and a second fitting: defining a second cavity; defining a second bore; comprising a second set of retention features arranged within the second cavity; and coupled to the mounting block; wherein the inner tube defines a distal tube section arranged within the second cavity of the second fitting; further comprising: a stopper: arranged within the distal tube section of the inner tube and extending through the second bore of the second fitting; and configured to block transfer of fluid within the inner tube to the mounting block; and a second ferrule: arranged about the distal tube section of the inner tube within the second cavity; and compressing the distal tube section of the inner tube, offset from a distal end of the inner tube, about the stopper; wherein the outer sleeve further defines a distal sleeve section defining: a second outer annular void between an inner wall of the second cavity and the outer sleeve; and a second inner annular void between the stopper and the outer sleeve; and further comprising a second volume of potting material arranged within the second cavity of the second fitting and configured to: occupy the second inner annular void and the second outer annular void within the second cavity of the second fitting; retain the distal tube section of the inner tube, the stopper, the second ferrule, and the distal sleeve section of the outer sleeve against the second fitting; and transfer tensile forces from the inner tube and the outer sleeve into the second fitting and the mounting block.

9. The system of claim 1: further comprising: a second fluid distribution block; and a second fitting: defining a second cavity; defining a second bore; comprising a second set of retention features arranged within the second cavity; and coupled to the second fluid distribution block; wherein the inner tube defines a distal tube section arranged within the second cavity of the second fitting; further comprising: a second needle: arranged within the distal tube section of the inner tube and extending through the second bore of the second fitting; and configured to transfer fluid from the second fluid distribution block through the inner tube to increase pressure within the inner tube; a second ferrule: arranged about the distal tube section of the inner tube within the second cavity; and compressing the distal tube section of the inner tube, offset from a distal end of the inner tube, about the second needle; wherein the outer sleeve further defines a distal sleeve section defining: a second outer annular void between an inner wall of the second cavity and the outer sleeve; and a second inner annular void between the second needle and the outer sleeve; and further comprising a second volume of potting material arranged within the second cavity of the second fitting and configured to: occupy the second inner annular void and the second outer annular void within the second cavity of the second fitting; retain the distal tube section of the inner tube, the second needle, the second ferrule, and the distal sleeve section of the outer sleeve against the second fitting; and transfer tensile forces from the inner tube and the outer sleeve into the second fitting and the second fluid distribution block.

10. The system of claim 1: wherein the inner tube defines a first durometer; and further comprising an intermediate tube; interposed between the inner tube and the outer sleeve; defining a second durometer greater than the first durometer; and configured to shield the inner tube from: extrusion through interstices between fibers in the outer sleeve; and abrasion by the outer sleeve resulting from inflation and deflation of the inner tube against the outer sleeve.

11. The system of claim 1: wherein the outer sleeve defines a first surface roughness; further comprising an intermediate sleeve: interposed between the proximal sleeve section of the outer sleeve and the inner wall of the first cavity of the first fitting; and defining a second surface roughness greater than the first surface roughness; and wherein the volume of potting material is configured to: impregnate interstices between fibers of the intermediate sleeve; bond the intermediate sleeve and the outer sleeve to the potting material; and retain the intermediate sleeve and the outer sleeve within the first cavity of the first fitting.

12. The system of claim 1, wherein the outer sleeve comprises a set of braided strands, the set of braided strands comprising: a first subset of strands: twisted in a first direction; and braided in a

second direction orthogonal to the first direction across the outer sleeve; and a second subset of strands: twisted in a third direction, opposite the first direction; braided in a fourth direction, orthogonal the third direction, across the outer sleeve; and cooperating with the first subset of strands to transfer tensile forces in an axial direction evenly across the outer sleeve toward the first fitting.

13. The system of claim 1, wherein the outer sleeve comprises a set of braided strands, each strand in the set of braided strands: formed of liquid crystal polymer fibers; and comprising an elastomeric coating configured to, responsive to increased pressure within the inner tube to expand the outer sleeve: retain the strands in a braided configuration across a length of the outer sleeve; and reduce friction between adjacent strands to increase flexibility of the outer sleeve.

14. The system of claim 1, wherein the first volume of potting material occupies the first outer annular void to form a radiused transition: extending from a distal end of first fitting to an outer surface of the outer sleeve; and configured to stress concentrations proximal the distal end of the first fitting during expansion of the outer sleeve.

15. The system of claim 1: wherein the first ferrule extends from the proximal tube section and terminates proximal a distal end of the first fitting to define a nominal gap between the outer sleeve and the inner tube; and wherein the inner tube is configured to, responsive to a first fluid pressure at the first fluid distribution block: expand to reduce the nominal gap between the outer sleeve and the inner tube; and expand the outer sleeve to an outer sleeve diameter, greater than the nominal sleeve diameter; and transfer tensile forces from the outer sleeve to the first fitting.

16. A system: comprising: a first fitting coupled to a first fluid distribution block; an inner tube defining a proximal tube section arranged within the first fitting; a first needle: arranged within the proximal tube section of the inner tube and extending through the first fitting; and configured to transfer a fluid from the first fluid distribution block into the inner tube to increase pressure within the inner tube; a first ferrule compressing the proximal tube section of the inner tube, offset from a proximal end of the inner tube, about the first needle; an outer sleeve defining a proximal sleeve section forming a void within the first fitting; and a first volume of potting material configured to: occupy the void within the first fitting; and retain the proximal tube section of the inner tube, the first needle, the first ferrule, and the proximal sleeve section of the outer sleeve against the first fitting; operable in a retracted position, the outer sleeve retracted to a nominal sleeve diameter responsive to supply of the fluid at a first fluid pressure to the inner tube by the first fluid distribution block, in the retracted position; and operable in an expanded position, the outer sleeve expanded to a first sleeve diameter, responsive to supply of the fluid at a second fluid pressure to the inner tube by the first fluid distribution block that expands the inner tube and the outer sleeve and increases tensile forces applied by the outer sleeve onto the first fluid distribution block, in the expanded position, the first sleeve diameter greater than the nominal sleeve diameter, and the second fluid pressure greater than the first fluid pressure.

17. The system of claim 16: further comprising an intermediate tube interposed between the inner tube and the outer sleeve, the intermediate tube offset from the proximal tube section of the inner tube; and wherein the first ferrule comprises a crimped ferrule comprising: an uncrimped section enclosing the intermediate tube and the inner tube; and a crimped section compressing the proximal tube section against an outer surface of the first needle to: form a seal configured to prevent flow of the fluid through an interstice between the outer surface of the first needle and the proximal tube section; and form a ramp of the proximal tube section over a proximal end of the crimped ferrule, the ramp configured to prevent flow of the fluid through the interstice.

18. The system of claim 16, further comprising the first fluid distribution block, the first fluid distribution block configured to transfer the fluid through the first needle into the inner tube, the fluid: flowing through a first interstice between the inner tube and an outer surface of the first needle; flowing through a second interstice between a first section of potting material, within the void, and the proximal tube section, the proximal tube section: forming a ramp over a proximal end

of the first ferrule from compression of the first ferrule, the first section of potting material, and the fluid against the proximal tube section; and configured to prevent flow of the fluid between the inner tube and the first ferrule; and flowing through a third interstice between an inner wall of the first fitting and a second section of potting material within the void, the inner wall comprising a first set of retention features configured to prevent flow of the fluid to a distal end of the first fitting.

19. The system of claim 16, wherein the outer sleeve comprises a set of braided strands, the set of braided strands comprising: a first subset of strands: twisted in a first direction; and braided in a second direction orthogonal to the first direction across the outer sleeve; and a second subset of strands: twisted in a third direction, opposite the first direction; braided in a fourth direction, orthogonal the third direction, across the outer sleeve; and cooperating with the first subset of strands to transfer tensile forces in an axial direction evenly across the outer sleeve toward the first fitting.

20. An apparatus comprising: a first fitting: an inner tube defining a proximal tube section arranged within the first fitting; a first needle: arranged within the proximal tube section of the inner tube and extending through the first fitting; and configured to transfer fluid into the inner tube to increase pressure within the inner tube; a first ferrule: arranged about the proximal tube section of the inner tube within the first fitting; and compressing the proximal tube section of the inner tube, offset from a proximal end of the inner tube, about the first needle; an outer sleeve: concentrically encompassing the inner tube; defining a nominal sleeve diameter configured to increase responsive to increased pressure within the inner tube; and defining a proximal sleeve section forming a void within the first fitting; and a first volume of potting material configured to: occupy the void within the first fitting; retain the proximal tube section of the inner tube, the first needle, the first ferrule, and the proximal sleeve section of the outer sleeve against the first fitting; and transfer tensile forces from the inner tube and the outer sleeve into the first fitting.
